

2

Statistical Inference

Syllabus

Need of statistics in Data Science and Big Data Analytics, Measures of Central Tendency : Mean, Median, Mode, Mid - range. Measures of Dispersion : Range, Variance, Mean Deviation, Standard Deviation. Bayes theorem, Basics and need of hypothesis and hypothesis testing, Pearson Correlation, Sample Hypothesis testing, Chi - Square Tests, t - test.

Contents

- 2.1 Need of Statistics in Data Science and Big Data Analytics
- 2.2 Measures of Central Tendency
- 2.3 Measures of Dispersion
- 2.4 Bayes Theorem
- 2.5 Hypothesis **Aug.-18, Oct.-19, Marks 5**
- 2.6 Pearson Correlation
- 2.7 Chi - square Tests
- 2.8 t - test **Aug.-18, May-19,**
..... **Oct.-19, Dec.-18, 19, Marks 6**
- 2.9 Multiple Choice Questions

2.1 Need of Statistics in Data Science and Big Data Analytics

- Statistics is the science of collecting, analyzing and understanding data and accounting for the relevant uncertainties. As such, it permeates the physical, natural and social sciences, public health, medicine, business and policy.
- As such, statistics is a fundamental tool of data scientists, who are expected to gather and analyze large amounts of structured and unstructured data and report on their findings.
- Statistics is a way to get information from data. Statistics is a tool for creating an understanding from a set of numbers. Statistics is the science of collecting, organizing, summarizing, analyzing, and interpreting information.
- Descriptive statistics involves computing values which summarize a set of data. This typically includes statistics like the mean, standard deviation, median, min, max, etc. which are called summary statistics.
- Fig. 2.1.1 shows descriptive statistics

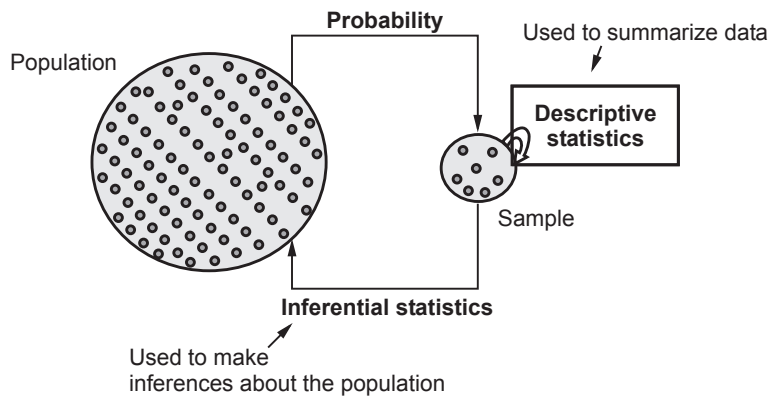


Fig. 2.1.1 Descriptive statistics

- Descriptive statistics is key because it allows us to present large amounts of raw data in a meaningful way. This enables a better interpretation of data.
- Descriptive statistics are also used to represent data graphically. Histograms, pie charts, bar graphs, and so on are nothing but a visual way of representing the data. This helps put it in the form that allows for analysis and interpretation.
- Big data is the collection and analysis of data sets that are complex in terms of the volume and variety, and in some cases the velocity at which they are collected. Big data are especially challenging because some of them were not collected to address a specific scientific question.

- Data scientists use a combination of statistical formulas and computer algorithms to notice patterns and trends within data. Then, they use their knowledge of social sciences and a particular industry or sector to interpret the meaning of those patterns and how they apply to real-world situations. The purpose is to generate value for a business or organization.
- While big data focuses on providing tools and techniques for managing and processing large and diverse quantities of data, it is not as focused on interpreting the data processing results to support decision making. This is where the area of data science continues and focuses on using advanced statistical techniques to analyze big data and interpret the results in a domain - specific context.
- Not everything is a table and not everything has a predefined schema. The massive data generation originating from social networks, mobile devices, sensors, and other data sources raised challenges that motivated the creation of novel tools and techniques.
- Initially, big data was characterized by 3 Vs : volume, variety, and velocity. Thus, enormous volumes of different and fast-growing data challenged the RDBMSs that do not scale easily due to ACID properties and fixed schema requirements.
- Many new Vs have emerged since then, such as data variability, veracity, value, etc. This caused the creation of new tools and frameworks that have the purpose of addressing one or more of the novel challenges.

2.2 Measures of Central Tendency

We look at various ways to measure the central tendency of data, include : Mean, Weighted mean, Trimmed mean, Median, Mode and Midrange.

1. Mean :

- The mean of a data set is the average of all the data values. The sample mean $\bar{x}$ is the point estimator of the population mean μ .

$$\text{Sample mean } \bar{x} = \frac{\text{Sum of the values of the } n \text{ observations}}{\text{Number of observations in the sample}} = \frac{\sum x_i}{n}$$

$$\text{Population mean } \mu = \frac{\text{Sum of the values of the } N \text{ observations}}{\text{Number of observations in the population}} = \frac{\sum x_i}{n}$$

2. Median :

- The median of a data set is the value in the middle when the data items are arranged in ascending order. Whenever a data set has extreme values, the median is the preferred measure of central location.

- The median is the measure of location most often reported for annual income and property value data. A few extremely large incomes of property values can inflate the mean.

- For an off number of observations :

7 observations = 26, 18, 27, 12, 14, 29, 19

Numbers in ascending order = 12, 14, 18, 19, 26, 27, 29

- The median is the middle value.

Median = 19

- For an even number of observations :

8 observations = 26, 18, 29, 12, 14, 27, 30, 19

Numbers in ascending order = 12, 14, 18, 19, 26, 27, 29, 30

The median is the average of the middle two values.

$$\text{Median} = \frac{(19+26)}{2} = 22.5$$

3. Mode :

- The mode of a data set is the value that occurs with greatest frequency. The greatest frequency can occur at two or more different values. If the data have exactly two modes, the data have exactly two modes, the data are bimodal. If the data have more than two modes, the data are multimodal.
- **Weighted mean :** Sometimes, each value in a set may be associated with a weight, the weights reflect the significance, importance, or occurrence frequency attached to their respective values.
- **Trimmed mean :** A major problem with the mean is its sensitivity to extreme (e.g., outlier) values. Even a small number of extreme values can corrupt the mean. The trimmed mean is the mean obtained after cutting off values at the high and low extremes.
- For example, we can sort the values and remove the top and bottom 2 % before computing the mean. We should avoid trimming too large a portion (such as 20 %) at both ends as this can result in the loss of valuable information.
- **Holistic measure** is a measure that must be computed on the entire data set as a whole. It cannot be computed by partitioning the given data into subsets and merging the values obtained for the measure in each subset.

2.3 Measures of Dispersion

- An **outlier** is an observation that lies an abnormal distance from other values in a random sample from a population.
- **First quartile (Q_1)** : The first quartile is the value, where 25 % of the values are smaller than Q_1 and 75 % are larger.
- **Third quartile (Q_3)** : The third quartile is the value, where 75 % of the values are smaller than Q_3 and 25 % are larger.
- The **box plot** is a useful graphical display for describing the behavior of the data in the middle as well as at the ends of the distributions. The box plot uses the median and the lower and upper quartiles. If the lower quartile is Q_1 and the upper quartile is Q_3 , then the difference ($Q_3 - Q_1$) is called the interquartile range or IQ.
- **Range** : Difference between highest and lowest observed values.

Variance :

- The variance is a measure of variability that utilizes all the data. It is based on the difference between the value of each observation (x_i) and the mean ($\bar{x}$) for a sample, μ for a population.
- The variance is the average of the squared between each data value and the mean.

$$\text{Sample variance : } S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{Population variance : } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Standard deviation :

- The standard deviation of a data set is the positive square root of the variance. It is measured in the same in the same units as the data, making it more easily interpreted than the variance.
- The standard deviation is computed as follows :

$$\text{Population standard deviation} = \sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

$$\text{Sample standard deviation} = S = \sqrt{S^2} = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Difference between standard deviation and variance

Sr. No.	Standard deviation	Variance
1.	Standard deviation is a measure of dispersion of the values of a data set from their mean.	It is the statistical measure of how far the numbers are spread in a data set from their average.
2.	It is a common term in statistical theory to calculate central tendency.	Variance is primarily used for statistical probability distribution to measure volatility from the mean.
3.	It measures the absolute variability of the dispersion.	It helps determine the size of the data spread.
4.	It is calculated by taking the square root of the variance.	It is calculated by taking the average of the squared deviation of each value in the data set from the mean.
5.	The standard deviation is symbolized by the Greek letter sigma " σ " as in lower case sigma.	The notation for the variance of a variable is " σ^2 " sigma squared.
6.	$\sigma = \sqrt{\frac{\sum (x - M)^2}{n}}$ where M = mean, x = a values in a data set and n = number of values	$\sigma^2 = \frac{\sum (x - M)^2}{n}$ where M = mean, x = each value in the data set, n = number of values in the data set
7.	Used in finance sector as a measure of market and security volatility.	Used in asset allocation.

Example 2.3.1 Find sample mean and sample standard deviation for the following data set : 5, 10, 15, 20.

Solution : Sample mean ($\bar{x}$) :

$$\bar{x} = \frac{\sum x}{n} = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4} = 12.5$$

Sample standard deviation :

Data	$x - \bar{x}$	$(x - \bar{x})^2$
5	$5 - (12.5) = -7.5$	$(-7.5)^2 = 56.25$
10	$10 - (12.5) = -2.5$	$(-2.5)^2 = 6.25$
15	$15 - (12.5) = 2.5$	$(2.5)^2 = 6.25$
20	$20 - (12.5) = 7.5$	$(7.5)^2 = 56.25$
		$\sum (x - \bar{x})^2 = 125.01$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{125.01}{4 - 1}} \approx 6.455$$

Example 2.3.2 The heights at the shoulders are : 600 mm, 470 mm, 170 mm, 430 mm and 300 mm. Find out the mean and the variance.

Solution :

Your first step is to find the Mean : $\frac{600 + 470 + 170 + 430 + 300}{5} = 394$

Variance :

$$600 - 394 = 206$$

$$470 - 394 = 76$$

$$170 - 394 = -224$$

$$430 - 394 = 36$$

$$300 - 394 = -94$$

$$\text{Variance} = \frac{(206)^2 + (76)^2 + (-224)^2 + (36)^2 + (-94)^2}{5}$$

$$\text{Variance} = \frac{42436 + 5776 + 50176 + 1296 + 8836}{5} = 21704$$

2.4 Bayes Theorem

- Bayes' theorem is a method to revise the probability of an event given additional information. Bayes's theorem calculates a conditional probability called a posterior or revised probability.
- Bayes' theorem is a result in probability theory that relates conditional probabilities. If A and B denote two events, $P(A|B)$ denotes the conditional probability of A occurring, given that B occurs. The two conditional probabilities $P(A|B)$ and $P(B|A)$ are in general different.
- Bayes' theorem gives a relation between $P(A|B)$ and $P(B|A)$. An important application of Bayes' theorem is that it gives a rule how to update or revise the strengths of evidence-based beliefs in light of new evidence a posteriori.
- A **prior probability** is an initial probability value originally obtained before any additional information is obtained.
- A **posterior probability** is a probability value that has been revised by using additional information that is later obtained.

- Suppose that $B_1, B_2, B_3, \dots, B_n$ partition the outcomes of an experiment and that A is another event. For any number, k , with $1 \leq k \leq n$, we have the formula :

$$P(B_k/A) = \frac{P(A/B_k) \cdot P(B_k)}{\sum_{i=1}^n P(A/B_i) \cdot P(B_i)}$$

Example 2.4.1 A mechanical factory production line is manufacturing bolts using three machines, A, B and C. The total output, machine A is responsible for 25 %, machine B for 35 % and machine C for the rest. The machines that 5 % of the output from machine A is defective, 4 % from machine B and 2 % from machine C. A bolt is chosen at random from the production line and found to be defective. What is the probability that it came from : i. machine A ii. machine B iii. machine C ?

Solution : Let,

$D = \{\text{bolt is defective}\},$

$A = \{\text{bolt is from machine A}\},$

$B = \{\text{bolt is from machine B}\},$

$C = \{\text{bolt is from machine C}\}.$

Given data : $P(A) = 0.25, P(B) = 0.35, P(C) = 0.4.$

$P(D|A) = 0.05, P(D|B) = 0.04, P(D|C) = 0.02.$

From the Bayes' Theorem :

$$\begin{aligned} P(A/D) &= \frac{P(D/A) \times P(A)}{P(D/A) \times P(A) + P(D/B) \times P(B) + P(D/C) \times P(C)} \\ &= \frac{0.05 \times 0.25}{0.05 \times 0.25 + 0.04 \times 0.35 + 0.02 \times 0.4} \\ &= \frac{0.0125}{0.0125 + 0.014 + 0.008} \end{aligned}$$

$$P(A/D) = 0.3621$$

Similarly :

$$\begin{aligned} P(B/D) &= \frac{P(D/B) \times P(B)}{P(D/A) \times P(A) + P(D/B) \times P(B) + P(D/C) \times P(C)} \\ &= \frac{0.04 \times 0.35}{0.05 \times 0.25 + 0.04 \times 0.35 + 0.02 \times 0.4} \end{aligned}$$

$$= \frac{0.014}{0.0125 + 0.014 + 0.008} = \frac{0.014}{0.0345}$$

$$P(B/D) = 0.4057$$

$$\begin{aligned} P(C/D) &= \frac{P(D/C) \times P(C)}{P(D/A) \times P(A) + P(D/B) \times P(B) + P(D/C) \times P(C)} \\ &= \frac{0.02 \times 0.4}{0.05 \times 0.25 + 0.04 \times 0.35 + 0.02 \times 0.4} \\ &= \frac{0.008}{0.0125 + 0.014 + 0.008} = \frac{0.008}{0.0345} \end{aligned}$$

$$P(C/D) = 0.2318$$

Example 2.4.2 *At a certain university, 4 % of men are over 6 feet tall and 1 % of women are over 6 feet tall. The total student population is divided in the ratio 3 : 2 in favour of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman ?*

Solution : Let us assume following :

M = {Student is Male},

F = {Student is Female},

T = {Student is over 6 feet tall}.

Given data : $P(M) = 2/5$,

$P(F) = 3/5$,

$P(T|M) = 4/100$

$P(T|F) = 1/100$.

We require to find $P(F|T)$?

Using Bayes' Theorem we have :

$$\begin{aligned} P(F/T) &= \frac{P(T/F) P(F)}{P(T/F) P(F) + P(T/M) P(M)} \\ &= \frac{\frac{1}{100} \times \frac{3}{5}}{\frac{1}{100} \times \frac{3}{5} + \frac{4}{100} \times \frac{2}{5}} = \frac{\frac{3}{500}}{\frac{3}{500} + \frac{8}{500}} \\ P(F/T) &= \frac{3}{11} \end{aligned}$$

Example 2.4.3 A pair of dice is rolled. If the sum of 9 has appeared, find the probability that one of the dice shows 3.

Solution : Let A = The event that the sum is 9

B = The event the one of dice shows 3.

Exhaustive cases = $6^2 = 36$.

Favorable cases of the event A = (3, 6), (6, 3), (4, 5), (5, 4).

So $P(A) = 4/36$

$$P(A) = \frac{1}{9}$$

Favorable case for the event $A \cap B = (3, 6), (6, 3)$

Hence $P(A \cap B) = \frac{2}{36} = \frac{1}{18}$

But $P(A \cap B) = P(A) \times P(B/A)$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(B/A) = \frac{1/18}{1/9} = \frac{1}{18} \times \frac{9}{1}$$

$$P(B/A) = 1/2$$

Example 2.4.4 At a certain university, 4 % of men are over 6 feet tall and 1 % of women are over 6 feet tall. The total student population is divided in the ratio 3 : 2 in favour of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman ?

Solution :

Let M = {Student is Male},

F = {Student is Female},

(note that M and F partition the sample space of students),

T = {Student is over 6 feet tall}.

We know that

$$P(M) = 2/5,$$

$$P(F) = 3/5,$$

$$P(T|M) = 4/100$$

$$P(T|F) = 1/100.$$

We require $P(F|T)$.

Using Bayes' Theorem we have :

$$P(F|T) = \frac{P(T/F)P(F)}{P(T/F)P(F) + P(T/M)P(M)} = \frac{\frac{1}{100} \times \frac{3}{5}}{\frac{1}{100} \times \frac{3}{5} + \frac{4}{100} \times \frac{2}{5}} = \frac{3}{11}$$

2.5 Hypothesis

SPPU : Aug.-18, Oct.-19

- General definition of a hypothesis : **"A hypothesis is a statement of a relationship between two or more variables."** A statistical hypothesis is simply a particular kind of hypothesis.
- A hypothesis is a statement or claim regarding a characteristic of one or more populations. Hypothesis testing is a procedure, based on sample evidence and probability, used to test claims regarding a characteristic of one or more populations.
- A statistical hypothesis is either
 1. A statement about the value of a population parameter (e.g., mean, median, mode, variance, standard deviation, proportion, total), or
 2. A statement about the kind of probability distribution that a certain variable obeys.
- The null hypothesis, denoted H_0 (read "H-naught"), is a statement to be tested. The null hypothesis is assumed true until evidence indicates otherwise. The alternative hypothesis, denoted, H_1 (read "H-one"), is a claim to be tested. We are trying to find evidence for the alternative hypothesis.
- Examples of statistical hypotheses :
 - a. The mean age of all college students is 20.4 years. **(simple hypothesis)**
 - b. The proportion of college students who are men is 60 %. **(simple hypothesis)**
 - c. The proportion of books in the college library whose heights exceed 30 cm is less than or equal to 0.13. **(composite hypothesis)**
- A statistical hypothesis that specifies a single value for a population parameter is called a simple hypothesis; every statistical hypothesis that is not simple is called composite.

2.5.1 Hypothesis Testing

- A statistical hypothesis test is a procedure for deciding between two possible statements about a population. The phrase significance test means the same thing as the phrase "hypothesis test."
- A hypothesis test is a statistical method that uses sample data to evaluate a hypothesis about a population. The general goal of a hypothesis test is to rule out chance as a plausible explanation for the results from a research study.
- The goal in hypothesis testing is to analyze a sample in an attempt to distinguish between population characteristics that are likely to occur and population characteristics that are **unlikely** to occur.

Basic assumption of hypothesis testing

- If the treatment has any effect, it is simply to add or subtract a constant amount to each individual's score.
- Remember that adding or subtracting constant changes the mean, but not the shape of the distribution for the population and/or the standard deviation.
- The population after treatment has the same shape and standard deviation as the population prior to treatment.
- If the individuals in the sample are noticeably different from the individuals in the original population, we have evidence that the treatment has an effect.

The purpose of the hypothesis test is to decide between two explanations :

1. The difference between the sample and the population can be explained by sampling error.
2. The difference between the sample and the population is too large to be explained by sampling error.

Steps in hypothesis testing

1. Specify the null hypothesis.
2. Specify the alternative hypothesis
3. Set the significance level (?)
4. Calculate the test statistic and corresponding P-value.
5. Display the conclusion.

Step 1 : Formulate the hypothesis

- A null hypothesis is a statement of the status quo, one of no difference or no effect. If the null hypothesis is not rejected, no changes will be made.
- An alternative hypothesis is one in which some difference or effect is expected.

- The null hypothesis refers to a specified value of the population parameter, not a sample statistic.

Step 2 : Select an appropriate test

- The **test statistic** measures how close the sample has come to the null hypothesis.
- The test statistic often follows a well-known distribution (e.g., normal, t, or chi-square).
- Calculate Z statistic.

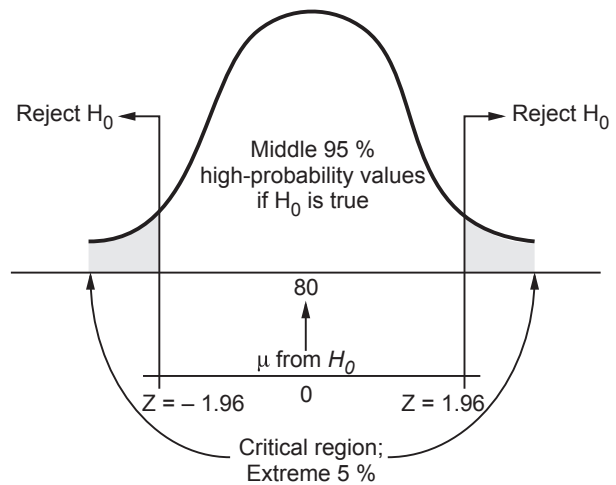


Fig. 2.5.1

Step 3 : Choose level of significance

Type I Error

- Occurs if the null hypothesis is rejected when it is in fact true.
- The probability of type I error (α) is also called the **level of significance**.

Type II Error

- Occurs if the null hypothesis is not rejected when it is in fact false.
- The probability of type II error is denoted by β .
- Unlike α , which is specified by the researcher, the magnitude of β depends on the actual value of the population parameter (proportion).
- **It is necessary to balance the two types of errors.**
- The power of a test is the probability $(1 - \beta)$ of rejecting the null hypothesis when it is false and should be rejected. Although β is unknown, it is related to α .

Step 4 : Collect data and calculate test statistic

- The required data are collected and the value of the test statistic computed. The test statistic z can be calculated as follows :

$$Z_{\text{cal}} = \frac{\hat{P} - \pi}{\sigma_P}$$

Step 5 : Determine probability value/critical value

- Using standard normal tables.
- Note, in determining the critical value of the test statistic, the area to the right of the critical value is either α or $\alpha/2$. It is α for a one-tail test and $\alpha/2$ for a two-tail test.

- If the prob associated with the calculated value of the test statistic (Z_{cal}) is less than the level of significance (α), the null hypothesis is rejected.
 - Alternatively, if the calculated value of the test statistic is greater than the critical value of the test statistic (z_{α}), the null hypothesis is rejected.
1. **Two-tailed alternative** : If the alternative states that a population parameter is different from a specific value. The corresponding test is called a two-tailed test.
 2. **Right-tailed alternative** : If the alternative states that a population parameter is greater than a specific value. The corresponding test is called a right-tailed test.
 3. **Left-tailed alternative** : If the alternative states that a population parameter is less than a specific value. The corresponding test is called a left-tailed test.

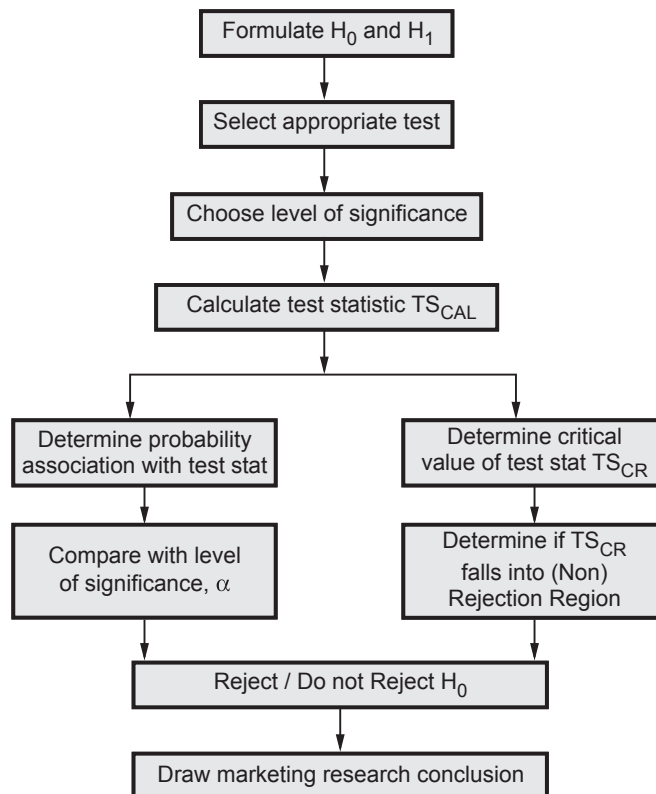


Fig. 2.5.2

Decide the rejection region of the test

- Based on the test statistic and a given confidence level, we can determine the rejection region, the acceptance region, and the critical value of the test.
- Rejection region is the region in which we can reject the null-hypothesis when the test statistics falls in this region. Acceptance region is simply the complement of the rejection region.

- Critical value is the value on the boundary of the rejection region and acceptance region.
- 1) For arbitrary population, acceptance and rejection regions are shown in Fig. 2.5.3.

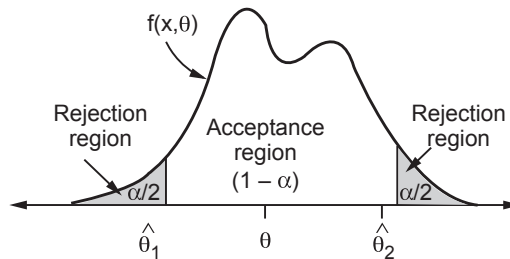


Fig. 2.5.3 Arbitrary population

- 2) For normal population, acceptance and rejection regions are shown in Fig. 2.5.4.

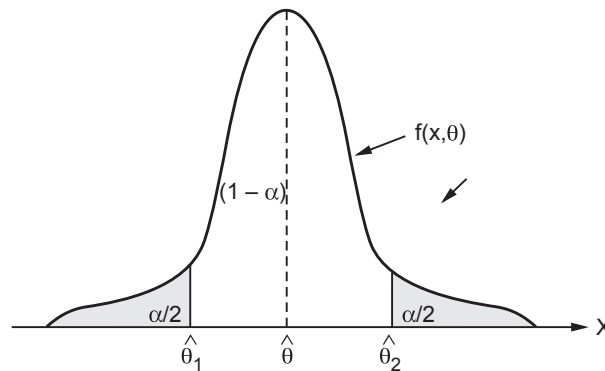


Fig. 2.5.4 Normal population

P-value and hypotheses testing

- As an alternative approach to the rejection/acceptance-region approach, we can calculate a probability related to the test statistic, called P-value, and base our decision of rejection/acceptance on the magnitude of the P-value.
- P-value is the probability to observe a value of the test statistic as extreme as the one observed, if the null hypothesis is true. So a small P-value indicates that the null hypothesis is not true and hence should be rejected.

In a hypothesis testing problem :

- a) The null hypothesis will not be rejected unless the data are not unusual (given that the hypothesis is true).
- b) The null hypothesis will not be rejected the P-value indicates the data are very unusual (given that the hypothesis is true).
- c) The null hypothesis will not be rejected only if the probability of observing the data provides convincing vidence that it is true.

- d) The null hypothesis is also called the research hypothesis; the alternative hypothesis often represents the status quo.
- e) The null hypothesis is the hypothesis that we would like to prove; the alternative hypothesis is also called the research hypothesis.

2.5.2 Null and Alternative Hypothesis

1. Null hypothesis (H_0)

- The null hypothesis states that there is no change in the general population before and after an intervention. In the context of an experiment, H_0 predicts that the independent variable had no effect on the dependent variable.
- The null hypothesis is the stated or assumed value of a population parameter. When trying to identify the population parameter needed for your solution, look for the following phrases :
 - i. "It is known that..."
 - ii. "Previous research shows..."
 - iii. "The company claims that..."
 - iv. "A survey showed that..."
- When writing the null hypothesis, make sure it includes an "=" symbol.

Null hypothesis : Population characteristic = Hypothesized value

2. Alternate hypothesis (H_1)

- The alternative hypothesis states that there is a change in the general population following an intervention. In the context of an experiment, predicts that the independent variable did have an effect on the dependent variable.
- The alternative hypothesis must be true when the null hypothesis is false.
- The alternate hypothesis is the stated or assumed value of a population parameter if the null hypothesis is rejected. When trying to identify the information needed for alternate hypothesis statement, look for the following phrases :
 - i. "Is it reasonable to conclude..."
 - ii. "Is there enough evidence to substantiate..."
 - iii "Does the evidence suggest..."
 - iv. "Has there been a significant..."
- When writing the alternate hypothesis, make sure it never includes an "=" symbol.
- **Alternative hypothesis :** Three possibilities
 - a. **Upper tailed test :** H_1 : Population characteristic > Hypothesised value
 - b. **Lower tailed test :** H_1 : Population characteristic < Hypothesised value
 - c. **Two tailed test :** H_1 : Population characteristic $\neq$ Hypothesised value

3. Multiple hypothesis testing :

- The multiple hypothesis testing problem is the situation when we wish to consider many hypotheses simultaneously. For example, suppose we have n genes and data about expression levels for each gene among healthy individuals and those with prostate cancer.
- However, consider a case where you have 20 hypotheses to test and a significance level of 0.05. What's the probability of observing at least one significant result just due to chance ?

$$\begin{aligned}
 P(\text{at least one significant result}) &= 1 - P(\text{no. significant results}) \\
 &= 1 - (1 - 0.05)^{20} \\
 &= 0.64
 \end{aligned}$$

- So, with 20 tests being considered, we have a 64 % chance of observing at least one significant result, even if all of the tests are actually not significant.
- In genomics and other biology-related fields, it's not unusual for the number of simultaneous tests to be quite a bit larger than 20 and the probability of getting a significant result simply due to chance keeps going up.

2.5.3 Difference between Null Hypothesis and Alternative Hypothesis

Sr. No.	Null hypothesis	Alternative hypothesis
1.	Represented by H_0 .	Represented by H_1 .
2.	Statement about the value of a population parameter.	Statement about the value of a population parameter that must be true if the null hypothesis is false.
3.	Always stated as an equality.	Stated in one of three forms : $>$, $<$, $\neq$
4.	This is the hypothesis or claim that is initially assumed to be true.	This is the hypothesis or claim which we initially assume to be false but which we may decide to accept if there is sufficient evidence.
5.	Independent variable had no effect on the dependent variable.	Independent variable did have an effect on the dependent variable.

Review Questions

1. Explain hypothesis testing with example.

SPPU : Aug.-18 (In Sem), Marks 4

2. Explain Type 1 and Type 2 errors.

SPPU : Aug.-18 (In Sem), Marks 2

3. Compare Type - I and Type - II errors.

SPPU : Oct.-19 (In Sem), Marks 5

4. Explain hypothetical testing in detail with example.

SPPU : Oct.-19 (In Sem), Marks 5

2.6 Pearson Correlation

- The Pearson correlation coefficient is used to measure the strength of a linear association between two variables, where the value $r = 1$ means a perfect positive correlation and the value $r = -1$ means a perfect negative correlation.
- Correlation means to find out the association between the two variables and Correlation coefficients are used to find out how strong the is relationship between the two variables. The most popular correlation coefficient is Pearson's correlation coefficient. It is very commonly used in linear regression.
- Pearson's Correlation coefficient is represented as 'r', it measures how strong is the linear association between two continuous variables using the formula :

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

where, r = Pearson correlation coefficient

x_i = x variable samples

y_i = y variable sample

$\bar{x}$ = mean of values in x variable

$\bar{y}$ = mean of values in y variable

- Fig. 2.6.1 shows various correlation with scatter diagram.

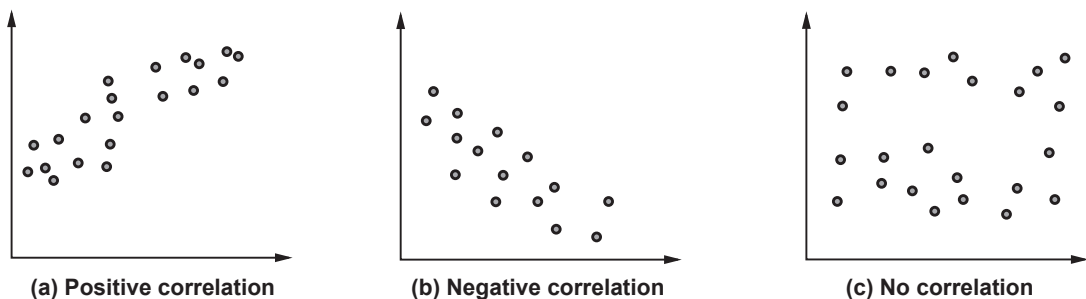


Fig. 2.6.1 Various correlation with scatter diagram

- Value of 'r' ranges from '- 1' to '+ 1'. Value '0' specifies that there is no relation between the two variables. A value greater than '0' indicates a positive relationship between two variables where an increase in the value of one variable increases the value of another variable. Value less than '0' indicates a negative relationship between two variables where an increase in the value of one decreases the value of another variable.

- Requirements for Pearson's correlation coefficient
 - a) Scale of measurement should be interval or ratio.
 - b) Variables should be approximately normally distributed.
 - c) The association should be linear.
 - d) There should be no outliers in the data.
- When an investigator has collected two series of observations and wishes to see whether there is a relationship between them, he or she should first construct a scatter diagram. The vertical scale represents one set of measurements and the horizontal scale the other.
- If one set of observations consists of experimental results and the other consists of a time scale or observed classification of some kind, it is usual to put the experimental results on the vertical axis. These represent what is called the "dependent variable". The "independent variable", such as time or height or some other observed classification, is measured along the horizontal axis, or baseline.

Example 2.6.1 Given below are the monthly income and their net savings of a sample of 10 supervisory staff belonging to a firm. Calculate the correlation coefficient.

Employee No.	1	2	3	4	5	6	7	8	9	10
Monthly income (₹)	780	360	980	250	750	820	900	620	650	390
Net savings	84	51	91	60	68	62	86	58	53	47

Solution : Following table list out some calculations.

Monthly income X	Net savings, y	X^2	Y^2	XY
780	84	608400	7056	65520
360	51	129600	2601	18360
980	91	960400	8281	89180
250	60	62500	3600	15000
750	68	562500	4624	51000
820	62	672400	3844	50840
900	86	810000	7396	77400
620	58	384400	3364	35960
650	53	422500	2809	34450

390	47	152100	2209	18330
$\sum X = 6500$	$\sum Y = 660$	$\sum X^2 = 4764800$	$\sum Y^2 = 45784$	$\sum XY = 456040$

Here, $n = 10$,

$$\bar{X} = \frac{\sum X}{n} = \frac{6500}{10} = 650$$

and
$$\bar{Y} = \frac{\sum Y}{n} = \frac{660}{10} = 66$$

Correlation coefficient is given as,

$$r(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} = \frac{\frac{1}{n} \sum XY - \bar{X}\bar{Y}}{\sqrt{\left(\frac{1}{n} \sum X^2 - \bar{X}^2\right) \left(\frac{1}{n} \sum Y^2 - \bar{Y}^2\right)}}$$

Putting values in above equation

$$r(X, Y) = \frac{\frac{456040}{10} - 650 \times 66}{\sqrt{\left(\frac{4764800}{10} - 650^2\right) \left(\frac{45784}{10} - 66^2\right)}} = 0.7804$$

Example 2.6.2 Following is the score of seven students in management accounting (X) and business statistics (Y). Calculate correlation between the score in two subjects.

Student no.	1	2	3	4	5	6	7
Score X	40	70	84	74	26	78	48
Score Y	64	74	100	60	50	48	80

Solution : Following table lists out some calculations.

X	Y	X ²	Y ²	XY
40	64	1600	4096	2560
70	74	4900	5476	5180
84	100	7056	10000	8400
74	60	5476	3600	4440

26	50	676	2500	1300
78	48	6084	2304	3744
48	80	2304	6400	3840
$\sum X = 420$	$\sum Y = 476$	$\sum X^2 = 28096$	$\sum Y^2 = 34376$	$\sum XY = 29464$

Here, $\bar{X} = \frac{\sum X}{n} = \frac{420}{7} = 60$

and $\bar{Y} = \frac{\sum Y}{n} = \frac{476}{7} = 68$

Correlation coefficient is given as,

$$r(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} = \frac{\frac{1}{n} \sum XY - \bar{X}\bar{Y}}{\sqrt{\left(\frac{1}{n} \sum X^2 - \bar{X}^2\right) \left(\frac{1}{n} \sum Y^2 - \bar{Y}^2\right)}}$$

Putting values in above equation

$$r(X, Y) = \frac{\frac{29464}{7} - 60 \times 68}{\sqrt{\left(\frac{28096}{7} - 60^2\right) \left(\frac{34376}{7} - 68^2\right)}} = 0.3749$$

Example 2.6.3 The competitors in a beauty contest are ranked by three judges in the following order. Use rank correlation coefficient to discuss which pair of judges has nearest approach to beauty.

1 st Judge	1	5	4	8	9	6	10	7	3	2
2 nd Judge	4	8	7	6	5	9	10	3	2	1
3 rd Judge	6	7	8	1	5	10	9	2	3	4

Solution : Here $n = 10$.

Let us form the following table of calculations.

1 st Judge x	2 nd Judge y	3 rd Judge z	d ₁ x - y	d ₂ y - z	d ₃ z - x	d ₁ ²	d ₂ ²	d ₃ ²
1	4	6	-3	-2	5	9	4	25
5	8	7	-3	1	2	9	1	4
4	7	8	-3	-1	4	9	1	16
8	6	1	2	5	-7	4	25	49
9	5	5	4	0	-4	16	0	16
6	9	10	-3	-1	4	9	1	16
10	10	9	0	1	-1	0	1	1
7	3	2	4	1	-5	16	1	25
3	2	3	1	-1	0	1	1	0
2	1	4	1	-3	2	1	9	4
			$\sum d_1 = 0$	$\sum d_2 = 0$	$\sum d_3 = 0$	$\sum d_1^2 = 74$	$\sum d_2^2 = 44$	$\sum d_3^2 = 156$

From above table, correlations are calculated as follows :

$$\rho(x, y) = 1 - \frac{6 \sum d_1^2}{n(n^2 - 1)} = 1 - \frac{6 \times 74}{10(100 - 1)} = 0.5515$$

$$\rho(y, z) = 1 - \frac{6 \sum d_2^2}{n(n^2 - 1)} = 1 - \frac{6 \times 44}{10(100 - 1)} = 0.7333$$

$$\rho(z, x) = 1 - \frac{6 \sum d_3^2}{n(n^2 - 1)} = 1 - \frac{6 \times 156}{10(100 - 1)} = 0.0545$$

Since $\rho(y, z)$ is maximum 2nd and 3rd judge have nearest approach to beauty.

2.7 Chi - square Tests

- There are many theoretical distributions, both continuous and discrete. We use 4 of these a lot : z (unit normal), t, chi-square and F.
- Z and t are closely related to the sampling distribution of means; chi-square and F are closely related to the sampling distribution of variances.

The definition of χ^2

- Scientists will often use the Chi-square (χ^2) test to determine the goodness of fit between theoretical and experimental data. In this test, we compare **observed values** with **theoretical or expected values**.
- Observed values are those that the researcher obtains empirically through direct observation; theoretical or expected values are developed on the basis of some hypothesis.
- The test statistic for comparing observed and expected frequencies is χ^2 defined as follows :

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\chi^2 = \sum_{i=1}^k \frac{(O-E)^2}{E}$$

- If V independent variables x_i are each normally distributed with mean μ_i and variance σ_i^2 then the quantity known as chi-square² is defined by

$$\begin{aligned}\chi^2 &= \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} + \dots + \frac{(x_v - \mu_v)^2}{\sigma_v^2} \\ &= \sum_{i=1}^v \frac{(x_i - \mu_i)^2}{\sigma_i^2}\end{aligned}$$

Where,

O = Observed value

E = Expected value

k = Number of categories, groupings, or possible outcomes

- Chi-square is the distribution of a sum of squares. Each squared deviation is taken from the unit normal : $N(0,1)$. The shape of the chi-square distribution depends on the number of squared deviates that are added together.

The critical values for the χ^2 distribution

- The use of the χ^2 distribution in hypothesis testing is analogous to the use of the t and F distributions. A null hypothesis is stated, a test statistic is computed, the observed value of the test statistic is compared to the critical value, and a decision is made whether or not to reject the null hypothesis.

2.7.1 Characteristics

1. It is not symmetric.
2. The shape of the chi-square distribution depends upon the degrees of freedom, just like Student's t-distribution.
3. As the number of degrees of freedom increases, the chi-square distribution becomes more symmetric as is illustrated in Fig. 2.7.1.
4. The values are non-negative. That is, the values are greater than or equal to 0.

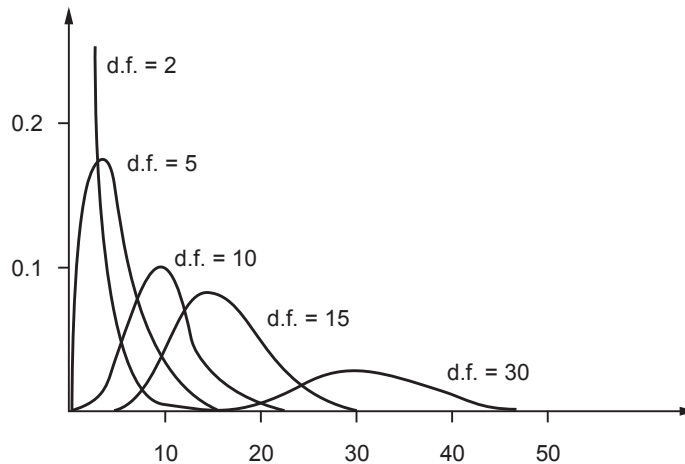


Fig. 2.7.1

- A **goodness-of-fit test** is an inferential procedure used to determine whether a frequency distribution follows a claimed distribution.
- The level of significance is used to determine the critical value. All chi-square goodness of fit tests are right tailed tests, so the critical value is with $k-1$ degree of freedom. In the Fig. 2.7.2, shaded region represents the critical region.
- The exact P-value for a non-directional test is the sum of probabilities for the table having a test statistic greater than or equal to the value of the observed test statistic.

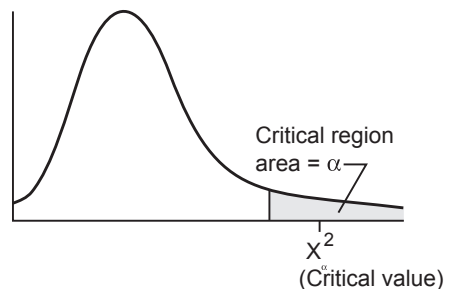


Fig. 2.7.2

1. High P-value : High probability that test statistic > Observed test statistic. **Do not reject** null hypothesis.
2. Low P-value : Low probability that test statistic > Observed test statistic. **Reject** null hypothesis.

Example 2.7.1 A firm manufacturing rivets wants to limit variations in their length as much as possible. The length (in cms) of 10 rivets manufactured by a new process are

2.15	1.99	2.05	2.12	2.17
2.01	1.98	2.03	2.25	1.93

Examine whether the new process can be considered superior to the old if the old population has standard deviation 0.145 cm ?

Solution : Given data : $n = 10$, $\sigma_0 = 0.145$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{2.15 + 1.99 + 2.05 + 2.12 + 2.17 + 2.01 + 1.98 + 2.03 + 2.25 + 1.93}{10}$$

$$\bar{x} = \frac{20.68}{10} = 2.068$$

$$S^2 = \frac{\sum (x - \bar{x})^2}{n}$$

x	$x - \bar{x}$	$(x - \bar{x})^2$
2.15	0.082	0.006724
1.99	- 0.078	0.006084
2.05	- 0.018	0.000324
2.12	0.052	0.002704
2.17	0.102	0.010404
2.01	- 0.058	0.003364
1.98	- 0.088	0.007744
2.03	- 0.038	0.001444
2.25	0.182	0.033124
1.93	- 0.138	0.019044
		0.09096

$$S^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{0.09096}{10} = 0.009096$$

Steps :

1. Null hypothesis (H_0) : $\sigma^2 = \sigma_0^2$
2. Alternative hypothesis (H_1) : $\sigma^2 > \sigma_0^2$
3. Level of significance $\alpha = 0.05$
4. Test statistic

$$\chi^2 = \frac{n s^2}{\sigma_0^2} = \frac{10 \times 0.009096}{(0.145)^2} = 4.326$$

Degree of freedom = $n - 1 = 10 - 1 = 9$

Tabulated χ^2 for 9 d.f. at 5 % level of significance is 16.919.

Calculated (4.326) < Tabulated (16.919)

Null hypothesis (H_0) is accepted.

New process cannot be considered superior to the old process.

2.7.2 Chi - square Test for Goodness of Fit

- The chi-square test is used to test if a sample of data came from a population with a specific distribution.
- An attractive feature of the chi-square goodness-of-fit test is that it can be applied to any uni-variate distribution for which you can calculate the cumulative distribution function.
- The chi-square goodness-of-fit test is applied to binned data.
- The chi-square goodness-of-fit test can be applied to discrete distributions such as the binomial and the Poisson.
- The chi square goodness of fit test begins by hypothesizing that the distribution of a variable behaves in a particular manner. For example, in order to determine daily staffing needs of a retail store, the manager may wish to know whether there are an equal number of customers each day of the week.
- To begin, an hypothesis of equal numbers of customers on each day could be assumed, and this would be the null hypothesis.
- Suppose that a variable has a frequency distribution with k categories into which the data has been grouped. The frequencies of occurrence of the variable, for each category of the variable, are called the observed values.
- The manner in which the chi square goodness of fit test works is to determine how many cases there would be in each category if the sample data were distributed exactly according to the claim.

- These are termed the expected number of cases for each category. The total of the expected number of cases is always made equal to the total of the observed number of cases.
 - The null hypothesis is that the observed number of cases in each category is exactly equal to the expected number of cases in each category.
 - The alternative hypothesis is that the observed and expected number of cases differ sufficiently to reject the null hypothesis.
 - Chi-square goodness of fit test is a non-parametric test that is used to find out how the observed value of a given phenomena is significantly different from the expected value.
 - In chi-square goodness of fit test, the term goodness of fit is used to compare the observed sample distribution with the expected probability distribution.
 - Chi-square goodness of fit test determines how well theoretical distribution (such as normal, binomial, or Poisson) fits the empirical distribution.
 - In chi-square goodness of fit test, sample data is divided into intervals. Then the numbers of points that fall into the interval are compared, with the expected numbers of points in each interval.
 - Let X be a random variable with unknown probability distribution function assuming the values $x_1, x_2, \dots$
 - Suppose that the values, say $x_1, x_2, \dots, x_i, \dots, x_n$ of size n , have occupied with frequencies $(Of)_1, (Of)_2, (Of)_3, \dots, (Of)_i, \dots, (Of)_n$ respectively, where (Of) stands for observed frequency and $\sum_{i=1}^n (Of)_i = N$.
1. State null hypothesized proportions for each category (p_i). Alternative is that at least one of the proportions is different than specified in the null.
 2. Calculate the expected counts for each cell as $n p_i$.
 3. Calculate the χ^2 statistic :

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

4. Compute the p-value as the proportion above the χ^2 statistic for either a randomization distribution or a χ^2 distribution with $df = (\text{number of categories} - 1)$ if expected counts all > 5
5. Interpret the p-value in context.

2.7.3 Chi - square Test for Independence of Attributes

- The chi-Square test of independence is used to determine if there is a significant relationship between two nominal (categorical) variables.

- The frequency of each category for one nominal variable is compared across the categories of the second nominal variable.
- The data can be displayed in a contingency table where each row represents a category for one variable and each column represents a category for the other variable.
- For example, say a researcher wants to examine the relationship between gender (male vs. female) and empathy (high vs. low). The chi-square test of independence can be used to examine this relationship.
- The null hypothesis for this test is that there is no relationship between gender and empathy.
- The alternative hypothesis is that there is a relationship between gender and empathy (e.g. there are more high-empathy females than high-empathy males).
- This test is also known as chi-square Test of Association. This test utilizes a contingency table to analyze the data.
- A contingency table is an arrangement in which data is classified according to two categorical variables. The categories for one variable appear in the rows, and the categories for the other variable appear in columns.
- Each variable must have two or more categories. Each cell reflects the total count of cases for a specific pair of categories.
- From the sample of N observations,
 1. $(Of)_{11}$ be the observed frequency of children of low income level going to government schools.
 2. $(Of)_{12}$ be the observed frequency of children of low income level going to private schools.
 3. $(Of)_{21}$ be the observed frequency of children of high income level going to government schools.
 4. $(Of)_{22}$ be the observed frequency of children of high income level going to private schools.
- This can be represented by 2×2 contingency table :

Income Level	Government	Private	Total
Low	$(Of)_{11}$	$(Of)_{12}$	R_1
High	$(Of)_{21}$	$(Of)_{22}$	R_2
Total	C_1	C_2	N

- Here R_1 and R_2 is row totals and C_1 and C_2 are column totals such that $C_1 + C_2 = R_1 + R_2 = N$, the total frequency.

	B_1	B_2	Row Total
A_1	a	b	$a + b \quad r_1$
A_2	$a c$	$b d$	$c + d \quad r_2$
Column Total	c_1	c_2	$a b c d = n$

- In case of 2×2 contingency table X^2 can be directly found using the short cut formula.

$$X^2 = \frac{n(ad - bc)^2}{c_1 \cdot c_2 \cdot r_1 \cdot r_2}$$

2.7.4 Strength and Limitation of Chi - square Test

Strength :

1. It is easier to compute than some statistics.
2. Chi-square makes no assumptions about the distribution of the population.

Limitations :

1. The chi-square test does not give us much information about the strength of the relationship or its substantive significance in the population.
2. The chi-square test is sensitive to sample size. The size of the calculated chi-square is directly proportional to the size of the sample, independent of the strength of the relationship between the variables.
3. The chi-square test is also sensitive to small expected frequencies in one or more of the cells in the table.

Example 2.7.2 If a sample of 900 units has mean of width of 3.5 cm and standard deviation of 2.61 cm then test whether this sample has come from a large population of mean width 3.25 cm and standard deviation 2.61 cm.

Solution : Given data :

Sample size (n) = 900 (large sample)

Sample mean ($\bar{x}$) = 3.50

Sample standard deviation (S) = 2.61

Population mean $\mu = \mu_0 = 3.25$ cm

Population standard deviation (σ) = 2.61 cm

Step 1 : Null hypothesis (H_0) : $\mu = \mu_0 = 3.25$

Step 2 : Alternative hypothesis (H_1) : $\mu \neq \mu_0 = 3.25$

Step 3 : Level of significance : No specific level of significance (α) is proposed. It can be assumed as either 5 % or 1 %.

When $\alpha = 5\%$ or $\alpha = 0.05$, we have for the two sided test

$$-Z_{\alpha/2} = -Z_{0.025} = -1.96 \text{ and } +Z_{\alpha/2} = +Z_{0.025} = +1.96$$

When $\alpha = 1\%$ or $\alpha = 0.01$, we have for the two sided test

$$-Z_{\alpha/2} = -Z_{0.005} = -2.58 \text{ and } +Z_{\alpha/2} = +Z_{0.005} = +2.58$$

Step 4 : Test statistics is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{3.50 - 3.25}{2.61/\sqrt{900}} = 2.87 = Z_2$$

Step 5 : Calculated Z value is greater than the tabulated Z values. So hypothesis is rejected at 5 % and 1 % level of significance.

Example 2.7.3 Four methods are under development for making discs of a superconducting material. 50 discs are made by each method, and they are checked for superconductivity when cooled with liquid .

	Method				
	M1	M2	M3	M4	Total
Superconductors	31	42	22	25	120
Failures	19	8	28	25	80
Total	50	50	50	50	200

Test the significance difference between the proportions of superconductors under different methods at 0.05 level using the chi-square test.

Solution :

Step 1 : Null hypothesis (H_0) : the proportions of semiconductors are equal.

Step 2 : Alternative hypothesis (H_1) : the proportions of semiconductors are not equal.

Step 3 : Computations of expected frequencies (E_f) is

Observed frequency (Of)	Expected frequencies (Ef)
(Of) ₁₁ = 31	(Ef) ₁₁ = (50 X 120) / 200 = 30
(Of) ₁₂ = 42	(Ef) ₁₂ = (50 X 120) / 200 = 30
(Of) ₁₃ = 22	(Ef) ₁₃ = (50 X 120) / 200 = 30
(Of) ₁₄ = 25	(Ef) ₁₄ = (50 X 120) / 200 = 30
(Of) ₂₁ = 19	(Ef) ₂₁ = (50 X 80) / 200 = 20
(Of) ₂₂ = 8	(Ef) ₂₂ = (50 X 80) / 200 = 20
(Of) ₂₃ = 28	(Ef) ₂₃ = (50 X 80) / 200 = 20
(Of) ₂₄ = 25	(Ef) ₂₄ = (50 X 80) / 200 = 20

Step 4 : Determination of degree of freedom = (m – 1) (n – 1) = (2 – 1) (4 – 1) = 3

Step 5 : Chi-square statistic

$$\begin{aligned}
 \chi^2 &= \frac{[(Of)_{ij} - (Ef)_{ij}]^2}{(Ef)_{ij}} \\
 &= \frac{(31-30)^2}{30} + \frac{(42-30)^2}{30} + \frac{(22-30)^2}{30} + \frac{(25-30)^2}{30} \\
 &\quad + \frac{(19-20)^2}{20} + \frac{(8-20)^2}{20} + \frac{(28-20)^2}{20} + \frac{(25-20)^2}{20} = 19.50
 \end{aligned}$$

Step 6 : Calculated chi-square test is greater than the tabulated value (7.815). We reject null hypothesis. Therefore, the proportions of superconductors are not equal.

2.8 t - test

SPPU : Aug.-18, May-19, Oct.-19, Dec.-18, 19

- Student's t-test assumes that distributions of the two populations have equal but unknown variances. A t-test is used as a hypothesis testing tool, which allows testing an assumption applicable to a population.
- The t-test is a parametric test of difference, meaning that it makes the same assumptions about your data as other parametric tests. The t-test assumes your data :
 - a) are independent
 - b) are (approximately) normally distributed.
 - c) have a similar amount of variance within each group being compared (a.k.a. homogeneity of variance)

- If your data do not fit these assumptions, you can try a nonparametric alternative to the t-test, such as the Wilcoxon Signed-Rank test for data with unequal variances.
- Suppose n_1 and n_2 samples are randomly and independently selected from two populations, pop1 and pop2, respectively.
- If each population is normally distributed with the same mean and with the same variance, then T (the t-statistic), given as

$$T = \frac{\bar{X}_1 - \bar{X}_2}{S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where,

$$S_P^2 = \frac{(n_1 - 1) S_1^2 + (n_2 - 1) S_2^2}{n_1 + n_2 - 2}$$

Properties of Student's t-Distribution :

1. The t-distribution is different for different degrees of freedom.
2. The t-distribution is centered at 0 and symmetric about 0.
3. The total area under the curve is 1. The area to the left of 0 is 1/2 and the area to the right of 0 is 1/2.
4. As the magnitude of t increases the graph approaches but never equals 0.
5. The area in the tails of the t-distribution is larger than the area in the tails of the normal distribution.
6. The shape of the t-distribution is dependent on the sample size n.
7. As sample size n increases, the distribution becomes approximately normal.
8. The standard deviation is greater than 1.
9. The mean, median and mode of the t-distribution are equal to zero.

2.8.1 Wilcoxon Rank - sum Test

- The Wilcoxon rank-sum test is a nonparametric alternative to the two sample t-test which is based solely on the order in which the observations from the two samples fall.
- It is used to test the null hypothesis that the median of a distribution is equal to some value.

- It can be used -
 - a) In place of a one-sample t-test
 - b) In place of a paired t-test
 - c) For ordered categorical data where a numerical scale is inappropriate but where it is possible to rank the observations.
- The logic behind the Wilcoxon test is quite simple. The data are ranked to produce two rank totals, one for each condition.
- If there is a systematic difference between the two conditions, then most of the high ranks will belong to one condition and most of the low ranks will belong to the other one.
- As a result, the rank totals will be quite different and one of the rank totals will be quite small.
- On the other hand, if the two conditions are similar, then high and low ranks will be distributed fairly evenly between the two conditions and the rank totals will be fairly similar and quite large.
- The Wilcoxon test statistic "W" is simply the smaller of the rank totals. The SMALLER it is taking into account how many participants you have then the less likely it is to have occurred by chance.
- A table of critical values of W shows you how likely it is to obtain your particular value of W purely by chance.
- There are N observations in all, where $N = n_1 + n_2$. Rank all N observations. The sum W of the ranks for the first sample is Wilcoxon rank sum statistic.
- If the two populations have the same continuous distribution, then W has mean,

$$\mu_w = \frac{n_1(N+1)}{2}$$

and standard deviation

$$\sigma_w = \sqrt{\frac{n_1 n_2 (N+1)}{2}}$$

- The **Wilcoxon rank sum test** rejects the hypothesis that the two populations have identical distributions when the rank sum W is far from its mean.

Example 2.8.1 Consider following data for wilcoxon rank-sum test :

Participant	Left ear	Right ear
1	25	32
2	29	30
3	10	7
4	31	36
5	27	20
6	24	32
7	27	26
8	29	33
9	30	32
10	32	32
11	20	30
12	5	32
median :	24.08	32.00

a) Find the difference between each pair of scores.

b) Rank these differences, ignoring any "0" differences and ignoring the sign of the difference (i.e. whether it is a positive or negative difference).

Solution :

Participant	Left ear	Right ear	Difference (d)
1	25	32	- 7
2	29	30	- 1
3	10	7	3
4	31	36	- 5
5	27	20	7
6	24	32	- 8
7	27	26	1
8	29	33	- 4

9	30	32	- 2
10	32	32	0
11	20	30	- 10
12	5	32	- 27

To rank the differences :

- Give the lowest rank to the smallest difference-score, ignoring whether it's a positive or negative difference.
- If two or more difference-scores are the same, this is a "tie": tied scores get the average of the ranks that those scores would have obtained, had they been different from each other.
- Here, ignoring the sign of the difference, the lowest difference is - 1.
- However there are two instances of this score (one positive and one negative).
- Therefore we add up the ranks that these scores would have had, if they had been different from each other (the ranks of 1 and 2), and then divide the sum of these ($1 + 2 = 3$) by the number of ranks involved (2).
- This gives us an "average" rank, 1.5, that we allocate to both of these two scores.
- The next lowest difference-score is - 2. We have now used up the ranks of 1 and 2, so this difference-score gets the ranks of 3.
- After that, ranking is straightforward until we get to the two difference scores of - 7 and 7. These would have got the ranks of 7 and 8, but instead get the average rank of 7.5 ($7 + 8 = 15$; $15/2 = 7.5$).
- This "uses up" the ranks of 7 and 8, so the next highest difference-score (- 8) gets the rank of 9.

Participant	Left ear	Right ear	Difference (d)	Ranked difference
1	25	32	- 7	7.5
2	29	30	- 1	1.5
3	10	7	3	4
4	31	36	- 5	6
5	27	20	7	7.5
6	24	32	- 8	9
7	27	26	1	1.5

8	29	33	- 4	5
9	30	32	- 2	3
10	32	32	0	ignore
11	20	30	- 10	10
12	0	32	- 27	11

- Add together the ranks belonging to scores with a positive sign (shaded in the table above) :

$$4 + 7.5 + 1.5 = 13$$

- Add together the ranks belonging to scores with a negative sign (unshaded in the table above) :

$$7.5 + 1.5 + 6 + 9 + 5 + 3 + 10 + 11 = 53$$

- Whichever of these sums is the smaller, is our value of W. So, $W = 13$.

Review Questions

1. Explain Wilcoxon rank sum test.

SPPU : Aug.-18 (In Sem), Marks 2, Oct.-19 (In Sem), Dec.-19 (End Sem), Marks 5

2. How Wilcoxon rank - sum test works ?

SPPU : May-19 (End Sem), Marks 5

3. When do we use Wilcoxon rank-sum test ? Write steps in the test.

SPPU : Dec.-18 (End Sem), Marks 6

2.9 Multiple Choice Questions

Q.1 The _____ of a data set is the average of all the data values.

- ☐ a mean
 ☐ b median
☐ c mode
 ☐ d all of these

Q.2 An _____ is an observation that lies an abnormal distance from other values in a random sample from a population.

- ☐ a range
 ☐ b quartiles
☐ c outlier
 ☐ d none

Q.3 The standard deviation of a data set is the positive square root of the _____.

- ☐ a sample
 ☐ b variance
☐ c population
 ☐ d outlier

Q.4 Pearson's correlation coefficient is represented by ____.

- ☐ a p ☐ b q
☐ c r ☐ d σ

Q.5 The Wilcoxon Rank Sum Test is often described as the ____ version of the two-sample t-test.

- ☐ a parametric ☐ b non-parametric
☐ c regular ☐ d none

Q.6 Chi-square test is a ____ method.

- ☐ a decision tree ☐ b clustering
☐ c hypothesis testing ☐ d correlation testing

Q.7 The mean, median and mode of the t-distribution are equal to ____.

- ☐ a zero ☐ b one
☐ c null ☐ d minus one

Q.8 In hypothesis testing, the hypothesis which is tentatively assumed to be true is called the ____.

- ☐ a correct hypothesis ☐ b null hypothesis
☐ c alternative hypothesis ☐ d level of significance

Answer Keys for Multiple Choice Questions :

Q.1	a	Q.2	c	Q.3	b
Q.4	c	Q.5	b	Q.6	c
Q.7	a	Q.8	b		



Notes

[illegible]